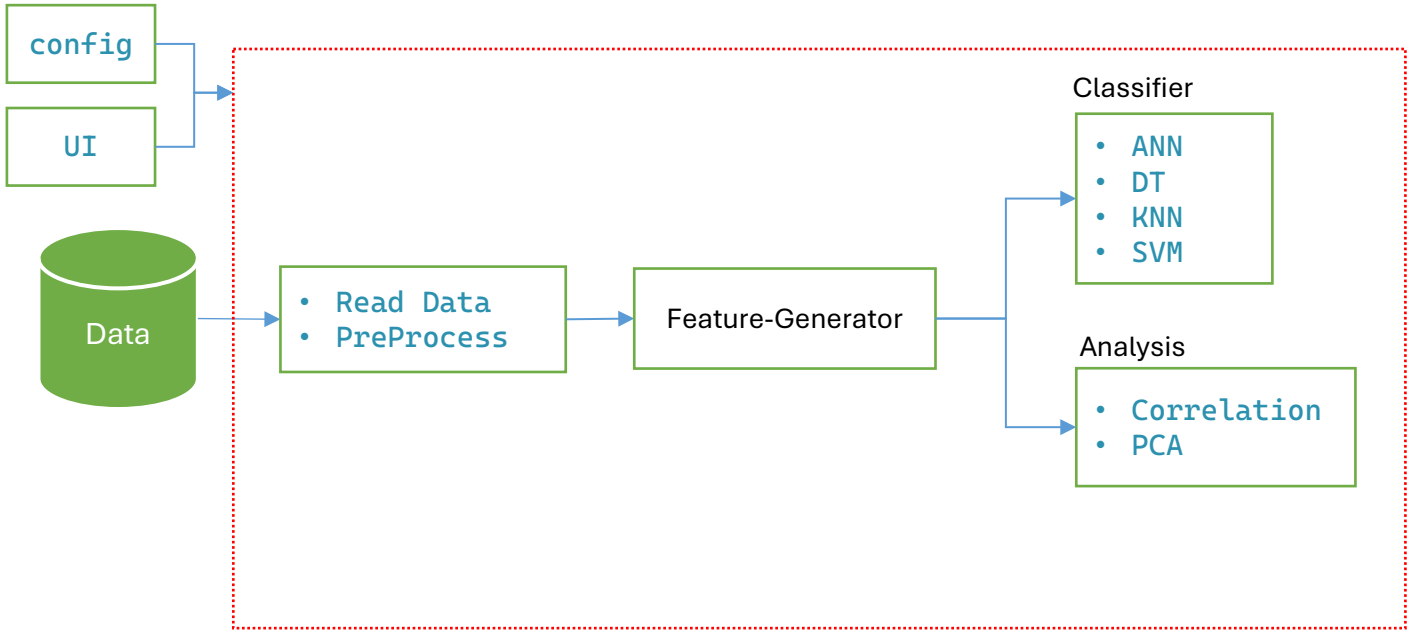
PROJECT REPORT
CMPS 470/570

TEAM:

TEAM MEMBERS & ROLES

MODULE COMMUNICATION GRAPH



- Export
  - Features
  - Model
  - Performance-epoch curve
  - Predictions
  - Performance scores
  - Plots
    - Performance-epoch curve
    - Performance scores

DESCRIPTION

- Goal: Identifying objects' class labels by means of ML algorithms ANN, SVM, DT, and K-NN.
- Tasks:
  - Visualize data
  - Preprocess data
  - Design representation
    - Extract features
    - Explore feature distributions
  - Design and create the model
  - Conduct model assessment

INSTRUCTIONS

- Develop a machine learning application using Python and related libraries that can
  - Pre-process data,
  - Create ML models based on the methods of k-NN, ANN, SVM, and DT.
  - Display/export results and model parameters
- Add “Team Member Participation” in your report.
- Submission
  - Submit project data at GitHub.
  - Submit your code and your PPT file at Canvas.

CODING STANDARDS

- Coding must follow readability requirements.
  - Use self-explanatory naming
  - Place proper comments that describes the logic
  - Divide the code into proper procedural sections such as {Pre-processing, Model, Assessment}.
- Must be efficient in terms of time and memory allocation
- User interface must allow to select the input data, export the model parameters, and export the test results.
- Use GitHub to share your code and the data set with your team member and your professor. Name your repository as
- "cmpsML_TEAM NAME".
  - Repository structure:
    - CODE
      - INPUT
      - OUTPUT
      - MODEL
    - DOC
    - OTHER
- Your program must
  - read from file(s) located in the folder INPUT\TRAIN and INPUT\TEST.
  - export outputs (test results, images, etc.) into the folder OUTPUT.
  - export model parameters into the folder MODEL.
  - be able to import model parameters and run the test at another time after creating the model.

CODING STANDARDS

1. Make the code generic.
2. Code must be efficient (data-structure, functionality).
3. Use a relative path.
4. Comment adequately.
    a) Add a comment for each code block, such as a loop-block, that describe the functionality
5. Always use secure data-types (such as immutables types).
6. Always test your code with an artificial data whose return value is known.
7. Make variables names self-explanatory.
8. Add the symbol # at the end of EACH block.
9. Do not calculate a common/constant value inside a loop.
10. Avoid initializing variables inside a loop unless it is required.
11. Initialize objects with None (null) (NOT zero) if their size is known instead of using append-like methods.
12. Save data in categorized folders.
13. Legends should be good enough in color, linestyle, shape etc. to distinguish data series.
14. Pay attention to the aesthetics.

DESCRIPTION OF THE PROJECT

DESCRIPTION OF THE RAW DATA

- Data source
- Attributes and their types
  - Use a table
- Distribution of values for each attributes
  - Use appropriate charts

PREPROCESSING

- Explain what you have done for pre-processing.
- Report parameters used in the pre-processing.
- Provide samples for pre-processed data.

FEATURE EXTRACTION & DESCRIPTION

- Explain what you have done for feature extraction.
- Report parameters used for feature extraction.
- Describe features
  - Data types, use a table
  - Use appropriate plots
  - Provide samples for features.

DESCRIPTION OF THE MODEL

- Describe parameters of the model.
- Use a search algorithm to find the best model parameters [MS Students]

PERFORMANCE OF THE MODEL

- *Plots*
  - Epoch-error curves for training and validation data sets on the same graph.
  - *Distribution of training, validation, test data sets*
  - *Performance measures to compare models*
- Performance measures
  - Apply the testing schemes including K-fold cross validation or 3-tier scheme (60, 20, 20).
    - MS students: Use both testing schemes.
  - Discuss performance of each model based upon the performance measures of sensitivity, specificity, accuracy, F1, AUC, and others as appropriate.
    - Plot them on the same figure
- Robustness of the models [MS Students]
  - Use the whisker-box plot out of at least 5 random trials